

Key Concepts and Terminologies

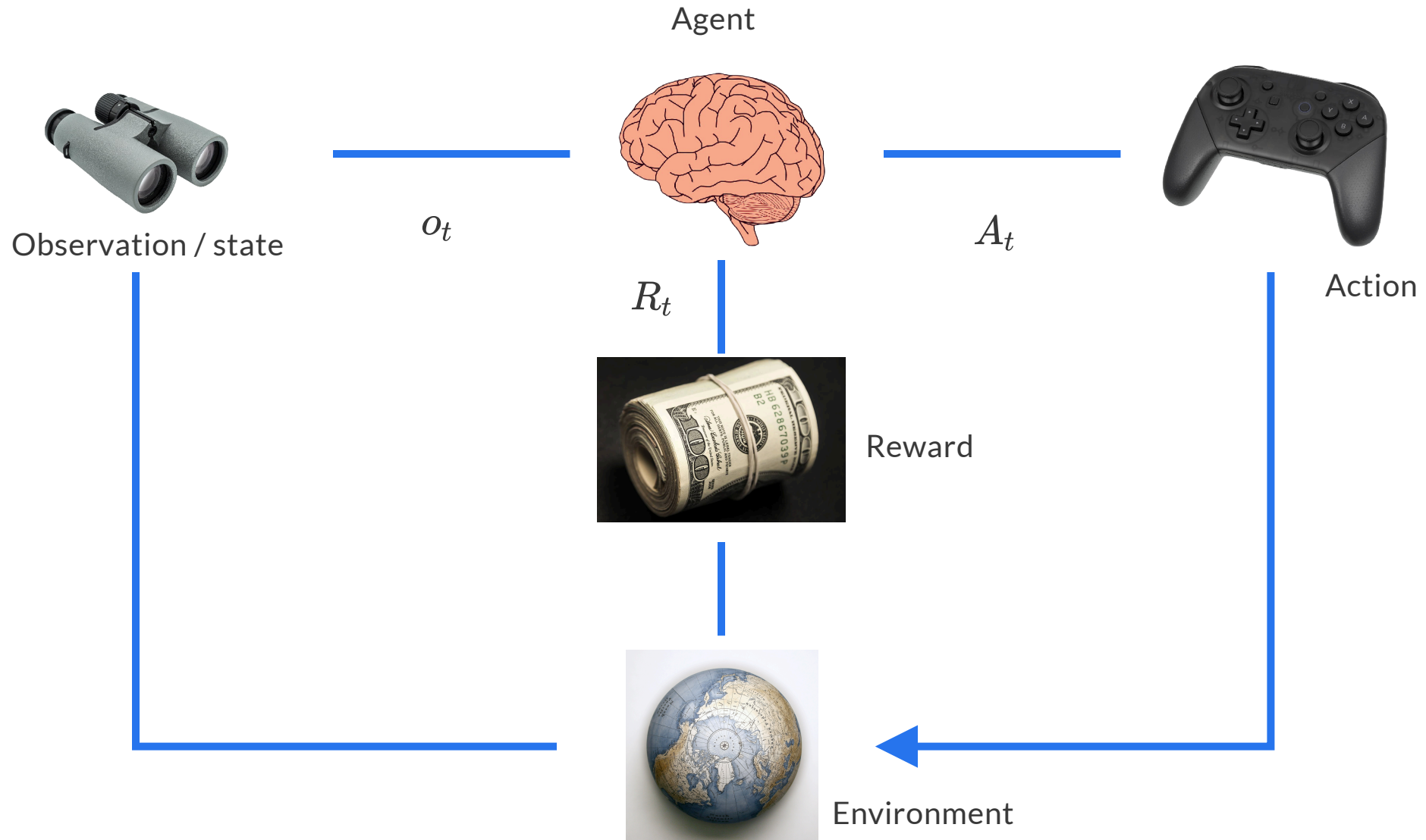
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Components in RL



Components of RL - Rewards

Reinforcement learning is based on reward hypothesis

1

Reward is a scalar feedback signal

2

Agents interact with environment and receive feedback

3

Reward indicates goodness of action by the agent

4

Objective is to maximize (discounted) cumulative reward

Reward Hypothesis (Sutton, 2004)

All of what we mean by goals and purposes can be well thought of as maximization of the expected value of the cumulative sum of a received scalar signal (reward).

Rewards: Examples

- Learning to bike

Positive reward for keeping the bike upright

Negative reward in case of a fall

- Portfolio maximization

Positive reward for profit

Negative reward for loss

- Autonomous navigation

Positive reward for maintaining safe distance from other vehicles

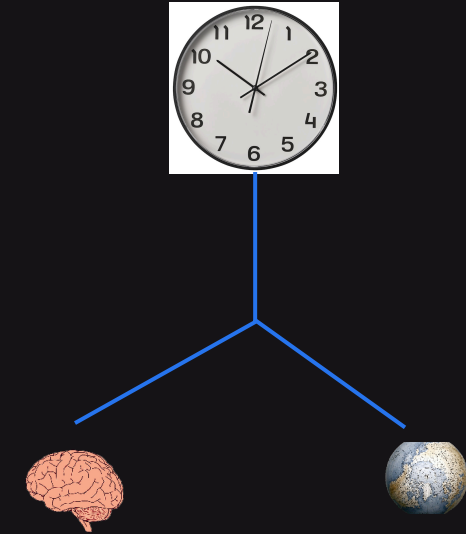
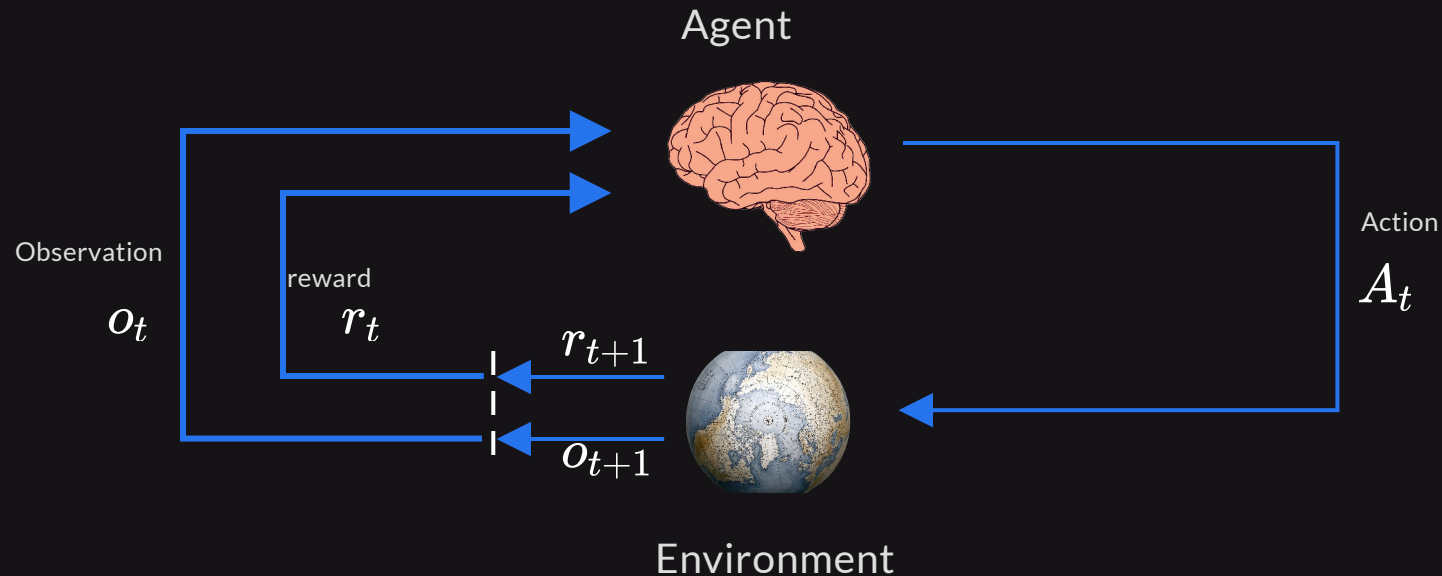
Negative reward for collision/aggressive maneuvering

- Power network management and control

Positive reward for managing uninterrupted power supply

Negative reward for blackouts

The Agent-Environment Interface



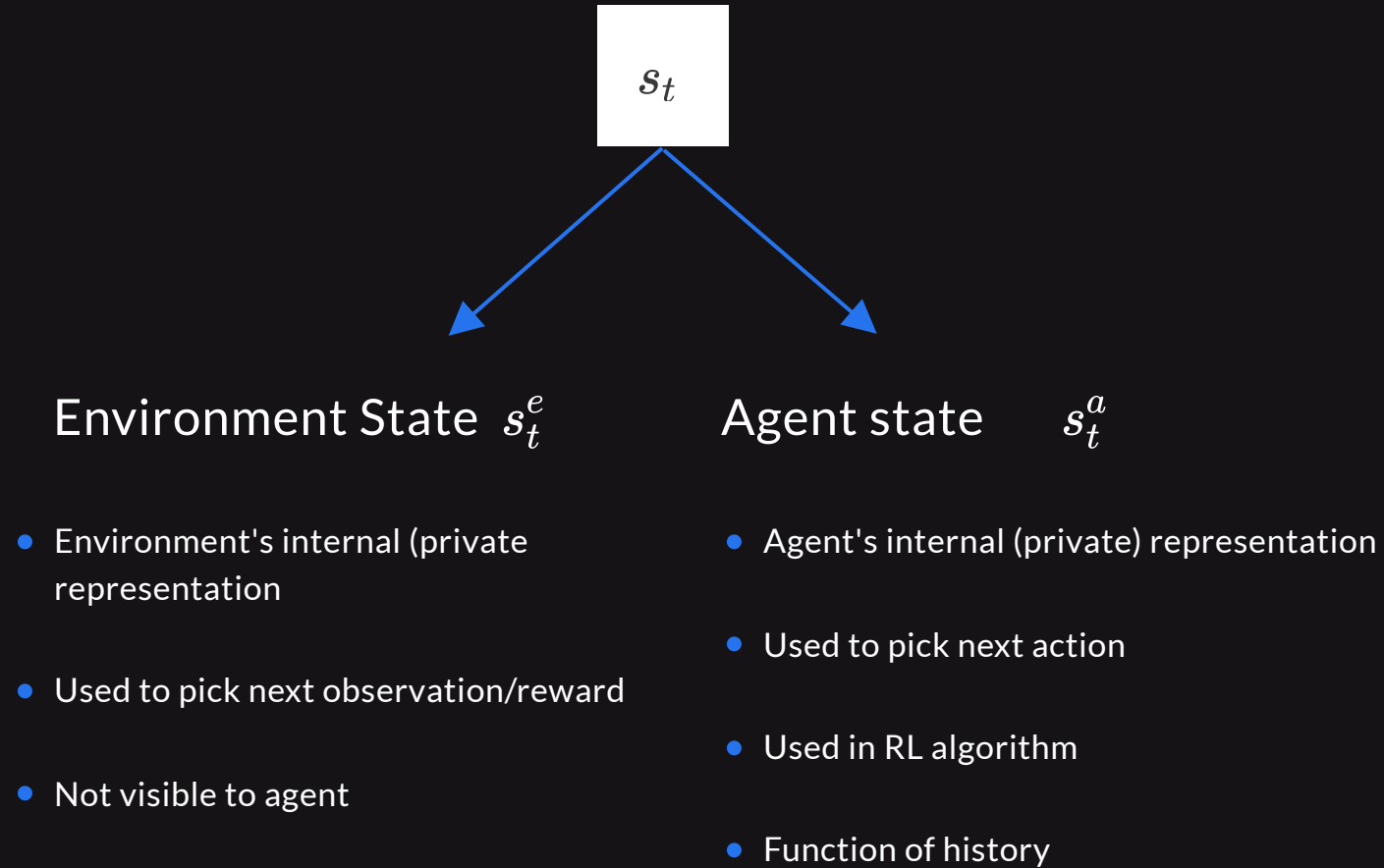
- Executes action A_t
- Receives observation O_t
- Receives reward R_t
- Receives action A_t
- Generates observation O_{t+1}
- Generates reward R_{t+1}

History and State



- All observable variables up to time, t
 - Sequence of observations, rewards, actions $(O_1, R_1, A_1, \dots, A_{t-1}, O_t, R_t)$
 - Future evolution depends on H_t
 - Agent selects action A_t
 - Environment selects
- State is a function of history: $s_t = f(H_t)$

History and State



History and State

Environment State

- Full Deck Composition

Complete set of cards remaining in the deck(s)

- Player's Cards

Cards in the player's hand

- Dealer's Cards

Both the dealer's face-up and face-down cards

- Other Player's Cards

Cards in the hands of other players (if applicable)

- Game Rules

Specific rules such as whether the dealer hits on a soft 17, etc.

- Betting Information

Current bets placed by all players

- History of Actions

Sequence of actions taken by all players and dealer in the current round

History and State

Agent State

- **Player's Cards**

The cards in the player's hand

- **Player's Total Bet**

The total amount of player has bet

- **Player's Usable Ace**

Whether the player has an ace that can be counted as 11 without busting

- **Other Player's Cards**

Cards in the hands of other players (if applicable)

- **Dealer's Visible Card**

The dealer's face-up card

- **Player's Current Hand Value**

The sum of the card values in the player's hand

- **Action History**

Player's previous actions in the current round (hit, stand, double down, split)

Thank You
